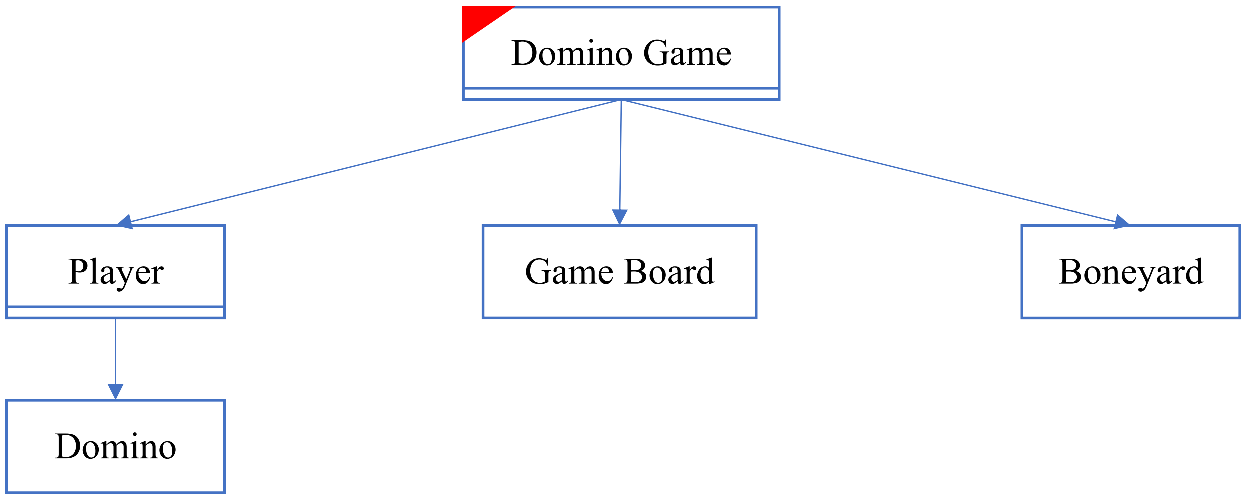
Proposed Design of Version 1



Trigger: update from keyboard

Description of the Design

Domino Game : Central manager of the game flow and state

- Data:
  - human: Represents the human player.
  - computer: Represents the computer player.
  - boneyard: The pool of undrawn dominos.
  - board: the current state of the game board
  - scanner: Scanner object for user input.
  - lastPlayer: Player who made the last move
- Behavior:
  - DominoGame(): Initialize players, boneyard, game board and scanner
  - play(): Start and manages the main game.
  - dealInitialHands(): Distributes initial dominos to players.
  - humanTurn(): Manages human player's turn.
  - computerTurn(): Manages computer player's turn.
  - playHumanDomino(): Process a human player's move
  - playComputerDomino(): Process a computer player's move
  - hasPlayableDomino(Player): Checks if a player has a playable domino.
  - canPlay(Domino): Checks if a domino can be played.
  - canPlayLeft(Domino): Checks if a domino can be played on the left end of the board
  - canPlayRight(Domino): Checks if a domino can be played on the right end of the board.

- isValidPlay(Domino, boolean): Checks if playing a domino is valid.
- isGameOver(): Checks if the game is over.
- displayGameState(): Displays the current state of the game.
- announceWinner(): Announces the winner of the game.

GameBoard: Represents the playing area where dominos are placed.

- Data:
  - playedDominos: List of Domino objects on the board.
  - leftEnd: int representing the value on the left end of the board.
  - rightEnd: int representing the value on the right end of the board.
- Behavior:
  - playDomino(Domino, boolean): Plays a domino to the board.
  - getLeftEnd(): Gets the value at the left end of the board.
  - getRightEnd(): Gets the value at the right end of the board.
  - toString(): Provides a string representation of the game board.

Player: Represents a player in the game.

- Data:
  - hand: The dominos in the player´s hand
  - name: String representing the player´s name.
- Behavior:
  - addDomino(Domino): Adds a domino to the player's hand.
  - getHand(): Gets the player's current hand.
  - getScore(): Calculates the player's score based on the sum of dominos in their hand.
  - removeDomino(Domino): Removes a domino from the player's hand.

Boneyard: Represents the pool of undrawn dominos.

- Data:
  - dominos: List of dominos in the boneyard.
- Behavior:
  - draw(): Draws a domino from the boneyard.
  - isEmpty(): Checks if the boneyard is empty.
  - size(): Gets the current size of the boneyard.

Domino: Represents a single domino piece.

- Data:
  - left: int representing the value on the left side.

    - right: int representing the value on the right side
- Behavior:
    - getLeft(): Gets the left value of the domino.
    - getRight(): Gets the right value of the domino.
    - rotate(): Rotates the domino by swapping its left and right values.
    - getSum(): Calculates the sum of the domino´s values
    - toString(): Provides a string representation of the domino.